

Task #4 Report

Note: as Mocha/Chai doesn't work with php, I had to adapt the web to html + js. Also, along the way I changed the functionality to fix some problems I had already detected, so it doesn't work exactly the same.

Task 4 Roman Numeral Converter — Automated Test Suite

Tests for `integerToRoman()` and `romanToInteger()` · Part 1: Human-created · Part 2: AI-assisted

Part 1 (Human-created)

`integerToRoman`

Invalid input

- ✓ TC-1 `integerToRoman(-2)` => Error
- ✓ TC-2 `integerToRoman(0)` => Error
- ✓ TC-19 `integerToRoman(4000)` => Error

Lower boundary

- ✓ TC-3 `integerToRoman(1)` => "I"

Subtractive-notation boundary values

- ✓ TC-4 `integerToRoman(4)` => "IV"
- ✓ TC-5 `integerToRoman(5)` => "V"
- ✓ TC-6 `integerToRoman(6)` => "VI"
- ✓ TC-7 `integerToRoman(9)` => "IX"
- ✓ TC-8 `integerToRoman(40)` => "XL"
- ✓ TC-9 `integerToRoman(49)` => "XLIX"
- ✓ TC-10 `integerToRoman(50)` => "L"
- ✓ TC-11 `integerToRoman(90)` => "XC"
- ✓ TC-12 `integerToRoman(100)` => "C"
- ✓ TC-13 `integerToRoman(400)` => "CD"
- ✓ TC-14 `integerToRoman(499)` => "CDXCIX"
- ✓ TC-15 `integerToRoman(500)` => "D"
- ✓ TC-16 `integerToRoman(900)` => "CM"
- ✓ TC-17 `integerToRoman(1000)` => "M"
- ✓ TC-18 `integerToRoman(3999)` => "MMMCMXCIX"

`romanToInteger`

Invalid characters

- ✓ TC-20 `romanToInteger("a")` => Error
- ✓ TC-21 `romanToInteger("AX")` => Error (invalid character A)
- ✓ TC-22 `romanToInteger("XA")` => Error (invalid character A)
- ✓ TC-23 `romanToInteger("XAX")` => Error (invalid character A)

Valid small values

- ✓ TC-24 `romanToInteger("I")` => 1
- ✓ TC-25 `romanToInteger("II")` => 2
- ✓ TC-26 `romanToInteger("III")` => 3

Repetition violations

- ✓ TC-27 `romanToInteger("IIII")` => Error (4 × I not allowed)
- ✓ TC-28 `romanToInteger("IIV")` => 4
- ✓ TC-29 `romanToInteger("VV")` => 5
- ✓ TC-30 `romanToInteger("VVV")` => Error (V cannot repeat)
- ✓ TC-31 `romanToInteger("VX")` => Error (V cannot subtract)
- ✓ TC-32 `romanToInteger("IX")` => 9
- ✓ TC-33 `romanToInteger("X")` => 10
- ✓ TC-34 `romanToInteger("XIII")` => Error (4 × I after X)
- ✓ TC-35 `romanToInteger("XL")` => 40
- ✓ TC-36 `romanToInteger("XXL")` => Error (invalid subtractive order)
- ✓ TC-37 `romanToInteger("IL")` => Error (I can only subtract from V, X)
- ✓ TC-38 `romanToInteger("L")` => 50
- ✓ TC-39 `romanToInteger("XCIX")` => 99
- ✓ TC-40 `romanToInteger("C")` => 100
- ✓ TC-41 `romanToInteger("CD")` => 400
- ✓ TC-42 `romanToInteger("CDXCIX")` => 499
- ✓ TC-43 `romanToInteger("D")` => 500
- ✓ TC-44 `romanToInteger("CM")` => 900
- ✓ TC-45 `romanToInteger("CMXCIX")` => 999
- ✓ TC-46 `romanToInteger("M")` => 1000
- ✓ TC-47 `romanToInteger("MMMCMXCIX")` => 3999
- ✓ TC-48 `romanToInteger("MMMM")` => Error (exceeds 3999)

Part 2 (AI-assisted)

integerToRoman

Invalid type inputs

- ✓ TC-A1 integerToRoman("III") => Error (string, not integer)
- ✓ TC-A2 integerToRoman(3.5) => Error (float)
- ✓ TC-A3 integerToRoman(null) => Error
- ✓ TC-A4 integerToRoman(undefined) => Error
- ✓ TC-A5 integerToRoman([]) => Error (array)
- ✓ TC-A6 integerToRoman(NaN) => Error
- ✓ TC-A7 integerToRoman(Infinity) => Error

Boundaries

- ✓ TC-A8 integerToRoman(1) => "I" (lower boundary)
- ✓ TC-A9 integerToRoman(3999) => "MMMCMXCIX" (upper boundary)
- ✓ TC-A10 integerToRoman(-1) => Error (just below lower boundary)
- ✓ TC-A11 integerToRoman(4000) => Error (just above upper boundary)

Subtractive-notation pairs

- ✓ TC-A12 integerToRoman(4) => "IV"
- ✓ TC-A13 integerToRoman(9) => "IX"
- ✓ TC-A14 integerToRoman(40) => "XL"
- ✓ TC-A15 integerToRoman(90) => "XC"
- ✓ TC-A16 integerToRoman(400) => "CD"
- ✓ TC-A17 integerToRoman(900) => "CM"

Mid-range representative values

- ✓ TC-A18 integerToRoman(14) => "XIV"
- ✓ TC-A19 integerToRoman(58) => "LVIII"
- ✓ TC-A20 integerToRoman(1994) => "MCMXCIV" (classic tricky case)
- ✓ TC-A21 integerToRoman(2024) => "MMXXIV"
- ✓ TC-A22 integerToRoman(2444) => "MMCDXLIV" (multiple subtractive pairs)
- ✓ TC-A23 integerToRoman(3888) => "MMMDCCCLXXXVIII" (longest valid numeral)

romanToInteger

Invalid non-string types

- ✓ TC-B1 romanToInteger(10) => Error (integer, not string)
- ✓ TC-B2 romanToInteger(null) => Error
- ✓ TC-B3 romanToInteger(undefined) => Error
- ✓ TC-B4 romanToInteger([]) => Error (array)
- ✓ TC-B5 romanToInteger("") => Error (empty string)

Invalid subtractive combinations

- ✓ TC-B6 romanToInteger("XIV") => Error (invalid subtractive combo)
- ✓ TC-B7 romanToInteger("LC") => Error (L cannot subtract)
- ✓ TC-B8 romanToInteger("DM") => Error (D cannot subtract)
- ✓ TC-B9 romanToInteger("IC") => Error (I can only subtract from V, X)
- ✓ TC-B10 romanToInteger("IM") => Error (I can only subtract from V, X)
- ✓ TC-B11 romanToInteger("XM") => Error (X can only subtract from L, C)
- ✓ TC-B12 romanToInteger("VL") => Error (V cannot subtract)

Whitespace violations

- ✓ TC-B13 romanToInteger(" X") => Error (leading whitespace)
- ✓ TC-B14 romanToInteger("X ") => Error (trailing whitespace)

Repetition violations

- ✓ TC-B15 romanToInteger("VV") => Error (V cannot repeat)
- ✓ TC-B16 romanToInteger("CCCC") => Error (C repeated 4 times)
- ✓ TC-B17 romanToInteger("LLLL") => Error (L cannot repeat)
- ✓ TC-B18 romanToInteger("DD") => Error (D cannot repeat)

Valid mid-range and complex values

- ✓ TC-B19 romanToInteger("XIV") => 14
- ✓ TC-B20 romanToInteger("LVIII") => 58
- ✓ TC-B21 romanToInteger("MCMXCIV") => 1994
- ✓ TC-B22 romanToInteger("MMXXIV") => 2024
- ✓ TC-B23 romanToInteger("MMMDCCCLXXXVIII") => 3888 (longest valid numeral)
- ✓ TC-B24 romanToInteger("MMCDXLIV") => 2444

Case sensitivity

- ✓ TC-B25 romanToInteger("x") => Error (lowercase not accepted)
- ✓ TC-B26 romanToInteger("xiv") => Error (all-lowercase not accepted)
- ✓ TC-B27 romanToInteger("Xiv") => Error (mixed-case not accepted)

All the proposed tests pass, and I don't know if it's due to the fixes I've done to the functionality or because the tests are insufficient.